

FOUNDATIONAL CONCEPTS OF ANALYTICS

1. Business Analytics is the practice of using data, statistical analysis, and modeling to make informed business decisions and solve problems. An example is a retail company analyzing its sales data to decide which products to stock more during holidays.

2. Data Analytics is the process of examining, cleaning, transforming, and modeling data to discover useful information, draw conclusions, and support decision making. An example is analyzing website traffic logs to understand user behavior and improve site navigation.

3. Business Intelligence is a set of technologies, applications, and practices for collecting, integrating, analyzing, and presenting business information to support better decision making. An example is using a BI tool like Tableau to create reports on monthly sales performance across regions.

4. Data Driven Decision making is making decisions based on data analysis and interpretation rather than intuition or observation alone. An example is a restaurant that uses customer feedback forms data to decide which menu items to remove or add.

5. Big Data refers to extremely large and complex data sets that require advanced tools and techniques to store, process and analyze. An example is social media platforms processing billions of user interactions daily to personalize content.

6. Descriptive Analytics is an analysis that summarizes historical data to understand what has happened in the past. Example is summarizing last quarter's sales by product category to see which performed best.

7. Diagnostic Analytics is an analysis focusing on understanding why something happened by identifying patterns and relationships in data. An example is investigating why sales dropped last month by analyzing customer complaints and market conditions.

8. Predictive Analytics is using statistical models and machine learning techniques to forecast future outcomes based on historical data.

an example is using past purchase data to forecast next month's demand for winter clothing.

1. Prescriptive Analytics is an advanced data analysis technique that evaluates multiple scenarios, constraints, and predictive models to recommend the optimal course of action. An example is a logistics company using optimization models to recommend the most cost effective delivery routes.

2. Data Literacy is the ability to read, understand, create, and communicate data as information. An example is training employees to interpret sales dashboards and make informed recommendations.

DATA, MEASUREMENT AND REPORTING

Structured data is a data that is organized in predefined formats, such as rows and columns in databases, making it easy to search and analyze. An example is customer information stored in a CRM system with fields like name, email, and purchase history.

Unstructured data is a data that does not have a predefined format, such as text, images, videos, and social media posts. An example is customer reviews and photos posted on social media platforms.

1. Data Warehouse is a centralized repository that stores integrated data from multiple sources for reporting and analysis. An example is a company like Woolworths that consolidates sales, marketing, and finance data from various departments into a central warehouse or headoffice for analysis.

4. Data mining is the process of discovering patterns and insights from large data sets using statistical and computational techniques. An example is when an online retailer identifies purchasing patterns to create targeted promotions.

3. Data Visualization is the graphical representation of data to help understand trends, outliers, and patterns. An example is creating a bar chart to compare monthly sales across different regions.

2. Dashboard is a visual interface that displays key metrics and performance indicators in a real time for quick insights. An example is a real-time dashboard showing website traffic, sales, and customer support tickets.

7. Metric is a quantifiable measure used to track and assess the status of a specific business process. Example is number of website visitors per day.

8. Key performance Indicator (KPI) is a measurable value that indicates how effectively a company is achieving key business objectives. Example: monthly revenue growth rate.

9. Benchmark is a standard or point of reference against which things can be compared or assessed. An example is comparing your company's customer satisfaction score against the industry average.

10. Segmentation is the process of dividing a broad customer or market base into sub-groups based on shared characteristics. Example: grouping customers by age and buying habits to tailor marketing messages.

CUSTOMER AND SALES KPI

1. Conversion Rate is the percentage of users who take a desired action, such as making a purchase or signing up. Example: 5% of website visitors complete a purchase.

2. Customer Acquisition Cost (CAC) is the total cost of acquiring a new customer, including marketing and sales expenses. An example: spending R2000 on ads to acquire new customers gives a CAC of R40.

3. Customer lifetime value (CLV/LTV) is the total revenue a business expects to earn from a customer over the entire duration of their relationship. Example is when a subscription service estimates the average customer will generate R500 in revenue over their lifetime.

4. Churn Rate is the percentage of customers who stop using a product or service during a specific period. Example: 10% of website subscribers cancel their service each month.

5. Customer Retention Rate is the percentage of customers who continue to use a product or service over a given period.

example of customer retention: 85% of customers renew their annual subscription.

16. Net Promoter Score (NPS) is a metric that measures customer loyalty by asking how likely they are to recommend a company to others. Example: customers rate their likelihood to recommend as 8/10, resulting in NPS of 50.

17. Average Order Value (AOV) is the average amount spent each time a customer places an order. Example: Total sales of R10 000 from 200 orders gives an AOV of R50.

18. Sales Funnel is a visual representation of the customer journey from initial awareness to final purchase. Example: from 1000 leads, 300 request demos, and 100 make purchases.

MARKETING AND CAMPAIGN ANALYTICS

19. Marketing Campaign is a coordinated series of marketing activities aimed at promoting a product, service or brand. Example is a campaign promoting a new smartphone with email, social media ads, and influencer partnerships.

20. Post Campaign Analysis is the evaluation of performance and impact of a marketing campaign after its completion. Example: reviewing sales and engagement data after a holiday promotion to measure success.

21. Incremental Revenue is the additional revenue generated as a direct result of a specific marketing campaign or activity. Example: A campaign drives an extra R50 000 in sales compared to the previous period.

22. Return on Investment (ROI) is a measure of the profitability of an investment, calculated as the ratio of net profit to the investment cost. Example: A R10 000 campaign generates R30 000 in profit, resulting in a 200% ROI.

23. Return on Ad Spend (ROAS) is a metric that measures the revenue generated for every money spent on advertising. Example: Spending R5000 on ads generates R25 000 in revenue, so ROAS is 5:1.

4. Click-Through Rate (CTR) is the percentage of people who click on an ad or link out of the total who viewed it. Example: 2% of people who saw an online ad clicked on it.

5. Bounce Rate is the percentage of visitors who leave a website after viewing only one page. Example: 40% of visitors leave a website without viewing other pages.

6. A/B Testing is a method of comparing two versions of a webpage or app to see which performs better. Example: testing two versions of a checkout page to see which results in more completed purchases.

7. Lead Generation is the process of attracting and converting prospects into potential customers. Example: collecting email addresses through a free ebook offer on a website.

8. Marketing Attribution is the process of identifying which marketing activities contribute to sales or conversions. Example: Assigning sales credit to the last clicked ad before purchase, or distributing credit among multiple touchpoints.

REFERENCES

Varish, V., Sharma, R., Rana, G., & Parashar, B. (2022). Foundation concepts for business analytics. In *Handbook of Business Analytics* (pp. 1-33). CRC Press.

Martini, R., Alirezai, G., Balda, E., & Benboodi, A. (2020). *Fundamentals of Data Analytics*. Springer International Publishing.

Warren, L. L., Owen, F. G., Albright, J. L., Appelman, R. D., Lamb, R. C., & Miller, J. D. (1977). Guidelines toward more uniformity in measuring and reporting self-experimental data. *Journal of Dairy Science*, 60(6), 1989-991.

Adams, P. S. (2007). Data analysis and reporting. In *Realtime PCR* (pp. 65-88). Taylor and Francis.

Westin, J. (2004). New Sales KPI's Reports to improve Data-based driven Decision making.

Reddy, J. R. S., Saitoja, B., & Reddy, M. N. (2024). Marketing campaign optimization. In *Predictive Analytics and Generative AI for Data-Driven Marketing Strategies* (pp. 114-120). Chapman and Hall/CRC.